AKSHIKA ROHATGI

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Curriculum Vitae as of May 2026

EDUCATION

The University of Texas at Austin 2023 – 2027 (expected)
Ph.D. in Computational Geophysics GPA 3.9/4.0
Advisors: Dr. Andrey Bakulin and Dr. Sergey Fomel

Indian Institute of Technology (IIT), Kharagpur 2019 – 2021
MS in Geophysics GPA 8.97/10.0, Institute Silver Medalist
Advisor: Dr. Sankar Kumar Nath

University of Delhi 2016 – 2019
BS (Hons) Physics GPA 8.39/10.0

RESEARCH EXPERIENCE

Texas Consortium for Computational Seismology (TCCS), UT Austin Aug 2023 – present
PIs: Dr. Sergey Fomel, Dr. Andrey Bakulin *Graduate Research Assistant*
- Developing algorithms to quantify and correct seismic phase perturbations for high resolution imaging.
- Developing and deploying a new Julia Package on local seismic attributes "LocalSignalAttribute"

University of Texas Institute of Geophysics (UTIG), UT Austin Apr 2021 – Sep 2021
PI: Dr. Mrinal K. Sen *Research Intern*
- Estimated seismic wavelets using statistical methods, constructed low-frequency models, and performed model-based post-stack inversion to generate seismic inversion images
- Segmented the obtained seismic inversion images using the K-means clustering algorithm.

Department of Geology and Geophysics, IIT Kharagpur Apr 2020 – May 2021
PI: Dr. Sankar Kumar Nath *Masters Thesis*
- Site Characterization and Seismic Hazard Analysis for 2001 Bhuj Earthquake in Kutch Region Gujrat, India.
- Achieved liquefaction susceptibility for the Bhuj 2001 earthquake.

CSIR – National Physical Laboratory, New Delhi Summer 2017, Summer 2019
PI: Dr. R.K. Kotnala *Research Intern*
- Developed biosensors using an impedance analyzer.

INDUSTRY EXPERIENCE

Chevron Center of Excellence May 2026 – Aug 2026
Research Intern Houston, Texas
- Implementing checkerboard to evaluate FWI resolvability of geologically interesting features under different acquisition and inversion scenarios.

TGS Research and Development May 2025 – Aug 2025
Research Intern Houston, Texas
- Developed the "Phase QC Tool" module in Imaging AnyWare software to analyze seismic phase.
- Applied phase despeckling on land data to enhance seismic resolution.

SLB Geophysics Technology (Formerly Schlumberger) Oct 2021 – Jun 2023
Commercialization Geophysicist Mumbai, India
- Worked on DELFI Seismic Processing platform for 3D seismic visualization.
- Deployed automated end-to-end tests using Protractor for CI/CD pipelines on Microsoft Azure.

Oil and Natural Gas Corporation May 2020 – Jun 2020
Intern Dehradun, India
- Worked on non-seismic methods of prospecting like gravity, electromagnetic, and magnetic methods.

INVITED TALKS

3. Invited Webinar Speaker, SLB SPI SIG Webinar, February 2026.

   Presented recent advances in applying circular statistics for seismic phase characterization, phase uncertainty analysis in seismic data.

2. Invited Seminar Speaker in Geophysical Society of Houston (GSH), February 2026.

   Discussed data-driven approaches for seismic phase analysis, including statistical modeling of phase behavior and applications to seismic signal enhancement.

1. AGU Fall Meeting 2024.

   Presented (on behalf of Dr. Andrey Bakulin) methodologies for designing geophysical monitoring workflows for subsurface hydrogen injection experiments.

PEER-REVIEWED PUBLICATIONS

6. A. Rohatgi, A. Bakulin, and S. Fomel, Application of seismic phase variance in processing QC. [*In Preperation*]

5. A. Rohatgi, A. Bakulin, Hassan Odhawani, and S. Fomel, A Quantitative Study Across Permian Basin Acquisitions. [*In Preperation*]

4. A. Rohatgi, A. Bakulin, and S. Fomel, Amplitude-invariant phase masking for coherence recovery in scattered wavefields. [*In Review*]

3. A. Rohatgi, A. Bakulin, and S. Fomel, Reducing local prestack waveform variability using circular-statistics phase masking. [*In Review*]

2. A. Rohatgi, A. Bakulin, and S. Fomel, Phase variance as a seismic quality-control attribute. [*In Review*]

1. A. Rohatgi, A. Bakulin, and S. Fomel, 2025, Data-driven analysis of seismic phase using circular statistics. *The Leading Edge (DOI: 10.1190/tle44090154.1)*.

CONFERENCE PROCEEDINGS

11. Rohatgi, A., Bakulin, A., Hassan, O. and Fomel, S., Quantifying offset-dependent seismic prestack data quality using data-driven metrics across different acquisition geometries. *International Meeting for Applied Geoscience & Energy, 2026, Expanded Abstract* [**Talk**]

10. Rohatgi, A., Bakulin, A., and Fomel, S., Diagnosing seismic prestack processing efficiency and bandwidth through frequency-dependent phase stability. *International Meeting for Applied Geoscience & Energy, 2026, Expanded Abstract* [**Talk**]

9. Rohatgi, A., Bakulin, A., Peterson, A., and Fomel, S., Reducing phase variability in prestack seismic using circular-statistics based phase masking. *International Meeting for Applied Geoscience & Energy, 2026, Expanded Abstract* [**Talk**]

8. Bakulin, A., Rohatgi, A., and Fomel, S., Taming Seismic Phase: From Statistical Characterization to Objective Phase Masking. *87th EAGE Annual Conference & Exhibition, 2026, Expanded Abstract* [**Talk**]

7. Rohatgi, A., Bakulin, A., and Fomel, S., Theory of seismic phase analysis using circular statistics. *International Meeting for Applied Geoscience & Energy, 2025, Expanded Abstract* [**Talk**]

6. Rohatgi, A., Bakulin, A., Fomel, S., and Badger. J., Impact of near-surface topography on reflection distortions: From diffractions to speckle noise. *International Meeting for Applied Geoscience & Energy, 2025, Expanded Abstract* [**Poster**]

5. Rohatgi, A., Bakulin, A., and Fomel, S., Seismic phase spectral analysis: Field-data insights from circular statistics. *International Meeting for Applied Geoscience & Energy, 2025, Expanded Abstract* [**Poster**]

4. Bakulin, A., Rohatgi, A., and Fomel, S., Statistical Analysis of Seismic Phase Variability in Dense Data. *86th EAGE Annual Conference & Exhibition, 2025, Expanded Abstract (DOI: 10.3997/2214-4609.2025101166)* [**Talk**]

3. Rohatgi, A., Bakulin, A., and Fomel, S., Analyzing the impact of additive and multiplicative noise in seismic data. *International Meeting for Applied Geoscience & Energy, 2024, Expanded Abstract (DOI: 10.1190/image2024-4086176.1)* [**Talk**]

2. Rohatgi, A., Bakulin, A., and Fomel, S., Phase Pilot Recovery for Mitigating Speckle Scattering Noise. *International Meeting for Applied Geoscience & Energy, 2024, Expanded Abstract (DOI: 10.1190/image2024-4091774.1)* [**Poster**]

1. Rohatgi, A., Bakulin, A., and Fomel, S., Seismic Speckle Noise: Recognizing Scattering Noise from Near-Surface Heterogeneities. *AGU Fall Meeting 2024 (DOI: 10.22541/essoar.173557561.19391758/v1)* [**E-lightning Talk**]

AWARDS & ACHIEVEMENTS

- Awarded the Exploration Geophysics Fellowship from the Jackson School of Geosciences, nominated by my advisor Dr. Sergey Fomel. Spring, 2026
- Winner of the JSG Geo-Hackathon on Computational Reproducibility, hosted by the Jackson School of Geosciences. [Github repo] 2024
- Recipient of the Great Start Scholarship at the Jackson School of Geosciences, The University of Texas at Austin. Fall, 2023
- Named Employee of the Month at Schlumberger Geophysics Technology Center (GTC) for contributions to geophysical technology development. Feb 2022
- Awarded the Institute Silver Medal at IIT Kharagpur, an academic honor given to students graduating with the highest academic performance in their department. 2021
- 99th percentile (All India Rank 384) in IIT-JAM Physics. 2019

TEACHING EXPERIENCE

Physical Geology (GEO 401) | Teaching Assistant | UT Austin Spring 2024
Led field sessions and taught introductory geology to a cohort of 80+ students.

TECHNICAL SKILLS

Software: Madagascar, Imaging AnyWare, Hampson Russell, Petrel, OMEGA, Delfi Zenith, MATLAB
Languages: Julia, Python, C, C++, MATLAB, SEGTeX, Typescript

SERVICE & VOLUNTEERING

- Session Co-Convener: Characterization of Natural Hydrogen Reservoir, IMAGE Conference 2025
- Manuscript Reviewer, Geophysics and The Leading Edge Journal 2025-present
- Abstract Reviewer, Society of Exploration Geophysicists (IMAGE Conference) 2025–present
- Newsletter Editor & Reviewer, SLB Geophysics Community 2022
- Executive Member, Prithvi Earth Science Fest, IIT Kharagpur 2019–2020

RELEVANT COURSES

UT Austin: Application of Machine Learning in Geosciences, Mathematics in Deep Learning, Seismic Data Processing, Introduction to Machine Learning in Geosciences, Computational Methods in Geosciences
IIT Kharagpur: Earthquake Seismology, Seismic Methods of Prospecting, Geophysical Inverse Theory, Geophysical Signal Processing, Borehole Geophysics, Numerical Modeling, Pattern Recognition in Geophysics
University of Delhi: Advanced Mathematical Physics, Wave Mechanics, Statistical Mechanics, Partial Differential Equations